

Database & Data Management

> Relevant source files

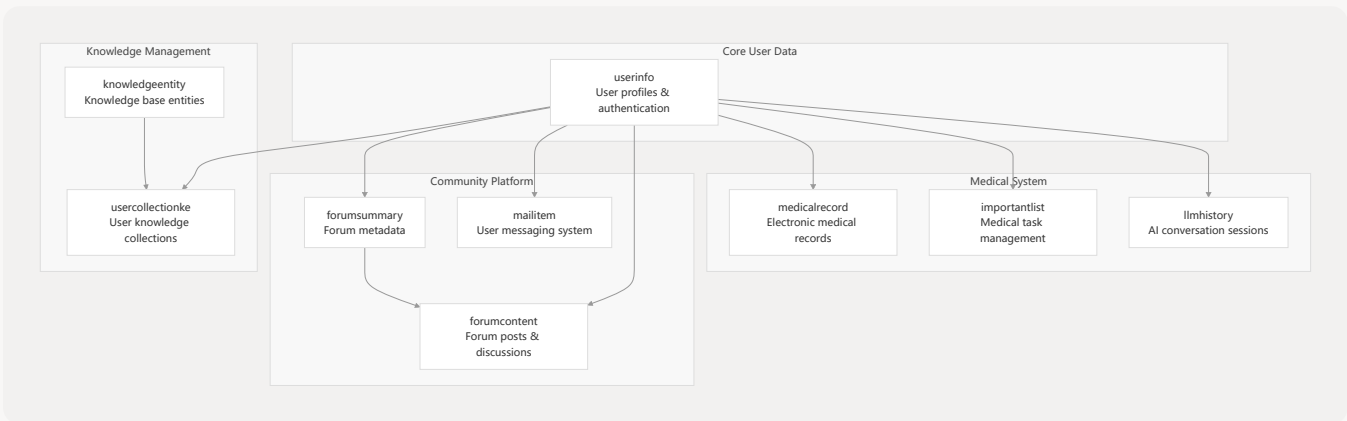
This document covers the MySQL database schema, data models, and storage architecture for the MKTY Smart Healthcare System. It details the relational database structure that supports user management, medical records, forum systems, knowledge entities, and AI conversation history.

For information about backend API data access patterns, see [Core API Endpoints](#). For administrative database management interfaces, see [Admin Backend API](#).

Database Schema Overview

The MKTY system uses a MySQL 8.0+ database named `mkty` with UTF-8 encoding. The schema consists of eight core tables that handle user data, medical records, forum content, knowledge management, and system interactions.

Database Architecture



Sources: `backend/script.sql` 1-188

Core Data Tables

User Information Table (`userinfo`)

The `userinfo` table serves as the central user registry with comprehensive profile management and privacy controls.

Field	Type	Purpose
<code>userId</code>	bigint	Primary key, auto-increment user identifier
<code>userCiphertext</code>	text	Encrypted password storage
<code>userName</code>	text	User display name
<code>userType</code>	text	User classification: 0=Patient, 1=Doctor, 2=Other
<code>userSex</code>	tinyint(1)	Gender: True=Male, False=Female
<code>userSexPermission</code>	tinyint(1)	Privacy setting for gender visibility
<code>userAge</code>	text	Age stored as text
<code>userAgePermission</code>	tinyint(1)	Privacy setting for age visibility
<code>userFrom</code>	text	Location/institution information
<code>userFromPermission</code>	tinyint(1)	Privacy setting for location visibility
<code>userContact</code>	text	Contact information (phone/email)
<code>userContactPermission</code>	tinyint(1)	Privacy setting for contact visibility
<code>patientImportantInfo</code>	text	Patient-specific medical history
<code>doctorImportantInfo</code>	text	Doctor specialization and department
<code>userImportantInfoPermission</code>	tinyint(1)	Privacy setting for important info
<code>userAvatarId</code>	text	GUID for user avatar image
<code>userRegisterTime</code>	bigint	Unix timestamp of registration

Sources: `backend/script.sql` 25-47

Medical Records Table (`medicalrecord`)

Electronic medical records linking patients with doctors and storing medical documentation.

Field	Type	Purpose
<code>medrecId</code>	bigint	Primary key for medical record
<code>medrecCreateTime</code>	text	Unix timestamp of creation
<code>medrecModifyTime</code>	text	Unix timestamp of last modification
<code>medrecPatientId</code>	bigint	Foreign key to patient's <code>userId</code>
<code>medrecDoctorId</code>	bigint	Foreign key to responsible doctor's <code>userId</code>

Field	Type	Purpose
medrecAbstract	text	Brief record summary
medrecState	text	Record status: 0=Active, 1=Recovered, 2=Chronic
medrecContent	text	Full record content in Markdown format

Sources: backend/script.sql 141-166

LLM History Table (llmhistory)

Stores AI conversation sessions and discussion mechanism data.

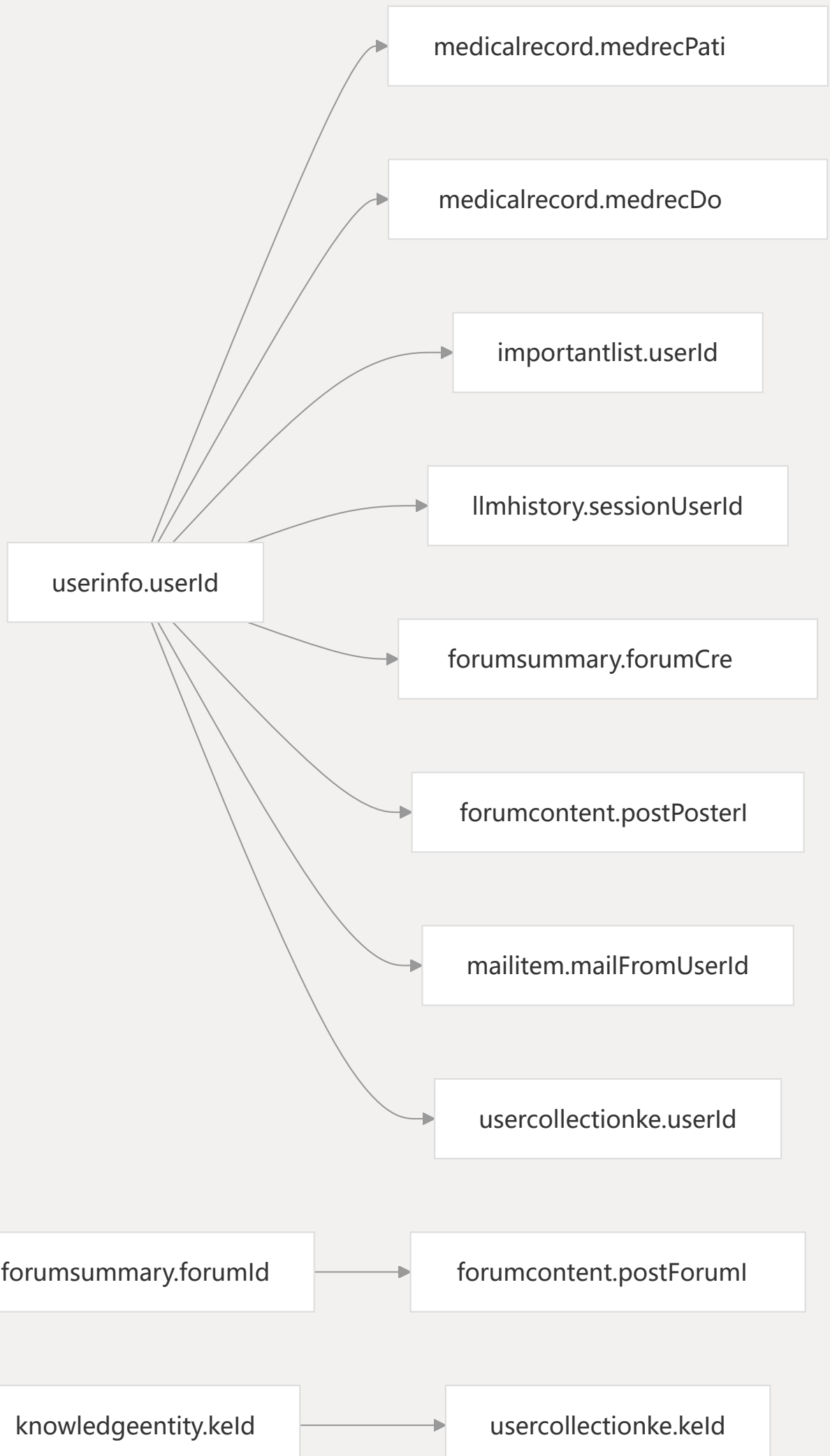
Field	Type	Purpose
sessionId	bigint	Primary key for conversation session
isSessionDM	tinyint(1)	Whether discussion mechanism is enabled
sessionSaveTime	text	Unix timestamp when session was saved
sessionUserId	bigint	Foreign key to user who owns session
sessionContent	json	Complete conversation data in JSON format

Sources: backend/script.sql 107-122

Data Relationships and Integrity

Foreign Key Constraints

The database enforces referential integrity through foreign key relationships:



Sources: backend/script.sql 59-178

Cascade Delete Behavior

Several tables implement cascade deletion to maintain data consistency:

- `forumcontent` cascades when referenced users or forums are deleted
- `llmhistory` cascades when the session owner is deleted
- `medicalrecord` cascades when either patient or doctor is deleted
- `usercollectionke` cascades when user or knowledge entity is deleted

Sources: backend/script.sql 75-178

Forum and Community Data Management

Forum Structure

The forum system uses a two-table design separating metadata from content:



Sources: `backend/script.sql` 49-87

JSON Content Storage

Forum posts use JSON storage for flexible content including images and text:

- `images` field: Array of GUID strings for WebP format images
- `content` field: Post text content string

Sources: `backend/script.sql` 71

Knowledge Entity and File Management

Knowledge Entity Table (`knowledgeentity`)

Manages knowledge base documents and files with GUID-based storage:

Field	Type	Purpose
keId	bigint	Primary key for knowledge entity
keGUID	text	GUID for file system directory naming
keFileType	text	MIME type of source file
keName	text	Display name/title
keCreateTime	text	Unix timestamp of upload
keAbstract	text	Entity summary/description

Sources: backend/script.sql 13-23

User Collections

The usercollectionke table creates many-to-many relationships between users and knowledge entities, enabling personal knowledge collections.

Sources: backend/script.sql 167-186

Task Management and Scheduling

Important List Table (importantlist)

Supports both one-time and recurring medical tasks: